
When Does Memory Improve Streaming RL?

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Abstract

Streaming reinforcement learning (RL) updates once per transition without replay or rollout batches. Under partial observability, recurrent state must be trained from the same local stream. We compare exact real-time recurrent learning (RTRL), one- and five-step truncated gradients, approximate e-prop, and delta-rule memory with recurrent PPO. RTU-RTRL solves Passive T-Maze through corridor length 10 and reaches 0.829 ± 0.297 success at length 15, but returns to the fixed-arm rate at lengths 30–50. On POPGym RepeatPrevious Easy at 170k transitions, PPO-GRU obtains 0.994 ± 0.001 return, RTU-RTRL 0.697 ± 0.036 , and TBPTT(5) 0.616 ± 0.036 . A bounded-radius RTU reaches 0.904 ± 0.063 in a separate 204.8k-transition configuration. Its target is perfectly linearly decodable despite negligible target-aligned clustering. Neither PPO nor streaming actor-critic adapts reliably in a five-seed switching-reward CT-graph. Exact online gradients help in a limited regime governed by recurrent stability, optimizer choice, and training budget.

1. Introduction

Modern deep RL commonly relies on replay, parallel environments, or multi-transition rollouts. A strict streaming agent instead consumes one transition, updates once, and discards it. Correlated targets and nonstationary gradients make deep networks unstable under this protocol, producing the *stream barrier* (Elsayed et al., 2024).

Partial observability adds a second difficulty: the policy must retain earlier observations and assign later rewards to the parameters that processed them. BPTT needs stored sequences; batch-size-one training usually retains only a one-transition recurrent gradient. Exact RTRL carries sensi-

tivities forward but is costly for dense recurrence (Williams & Zipser, 1989). Recurrent trace units (RTUs) make the exact trace tractable through diagonal dynamics (Elelimy et al., 2024; Farr et al., 2026). Whether that exact gradient improves control relative to recurrent PPO remains unclear (Schulman et al., 2017; Ni et al., 2022).

We evaluate CartPole, Passive and Active T-Maze, and POPGym (Towers et al., 2024; Ni et al., 2022; Morad et al., 2023). The recurrent models include GRUs (Cho et al., 2014), LSTMs (Hochreiter & Schmidhuber, 1997), RTUs, approximate e-prop, a delta-rule cell, and fixed memory. Matched PPO uses one environment but retains rollout storage and repeated epochs.

The paper makes three contributions:

- A matched study across CartPole, T-Maze, and POPGym measures when RTU-RTRL improves over one-step training and how far it remains behind recurrent PPO.
- An implementation audit corrects nominal TBPTT(5) and e-prop paths that did not use their intended temporal gradients.
- Frozen-policy probes and a five-seed switching-reward experiment separate stored information, policy use, and continual adaptation.

2. Related Work

Streaming deep RL. Stream-X combines eligibility traces, online normalization, sparse initialization, layer normalization, and overshooting-bounded gradient descent (ObGD) (Elsayed et al., 2024). Its Appendix B adds RMSProp-style variance normalization before the ObGD bound. Adaptive $Q(\lambda)$ instead normalizes traces, clips TD errors, and removes that bound (Gogianu et al., 2026). We use ordinary ObGD for CartPole, T-Maze, stable RepeatPrevious, and CT-graph; the matched 170k RepeatPrevious study uses bias-corrected AdaptiveObGD. PPO uses Adam.

Memory under partial observability. Tuned recurrent model-free RL is competitive across many POMDPs (Ni et al., 2022). POPGym isolates algorithmic memory demands (Morad et al., 2023). Memory and credit-assignment

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length can differ substantially (Ni et al., 2023); we therefore report gradient fidelity, state decodability, and policy return separately.

Online recurrent credit assignment. RTRL propagates exact forward sensitivities but scales poorly for dense recurrence (Williams & Zipser, 1989). RTUs replace the dense recurrent matrix with a diagonal transition and admit a compact exact trace (Elelimy et al., 2024); recent work combines them with streaming actor-critic (Farr et al., 2026). E-prop factors gradients into local traces and learning signals (Bellec et al., 2020). We compare these estimators under the same online update.

Fast-weight and lifelong memory. DeltaNet replaces additive fast-weight writes with a correction to the value stored at a key (Schlag et al., 2021); Gated DeltaNet adds data-dependent forgetting (Yang et al., 2025). We test a smaller recurrent variant. CT-graph supplies controlled reward switches for continual evaluation (Soltoggio et al., 2023). Stochastic stateful policy gradients train memory without BPTT (Al-Hafez et al., 2023), while frozen memory trains only a readout (Paischer et al., 2022); PPO-RS and a frozen SSM serve as controls.

3. Methods

3.1. Streaming Setting and Actor-Critic

We consider a POMDP with observation o_t , action a_t , reward r_t , and recurrent state h_t . A strict streaming learner updates once from (o_t, a_t, r_t, o_{t+1}) without replay or rollout storage:

$$h_t = f_\theta(h_{t-1}, o_t), \quad a_t \sim \pi_\theta(\cdot \mid h_t, o_t). \quad (1)$$

Actor and critic states are separate, and one environment is active unless stated otherwise. We distinguish memory retention from credit assignment: a probe can decode an old cue even when later rewards cannot train the parameters that processed it. Our base method is StreamAC(λ). With TD error

$$\delta_t = r_t + \gamma V_w(o_{t+1}, h_{t+1}) - V_w(o_t, h_t), \quad (2)$$

the actor eligibility trace and update are

$$e_t^\pi = \gamma \lambda e_{t-1}^\pi + \nabla_\theta \log \pi_\theta(a_t \mid o_t, h_t), \quad (3)$$

$$\Delta \theta_t \propto \delta_t e_t^\pi + \eta \nabla_\theta \mathcal{H}(\pi_\theta). \quad (4)$$

The critic uses the analogous value-gradient trace. Online Welford statistics normalize observations and rewards; LayerNorm, leaky ReLU, 90% sparse initialization, and ObGD follow Stream-X (Elsayed et al., 2024). For trace e_t , ObGD uses

$$\alpha_t = \frac{\alpha}{\max(1, \alpha \kappa \max(|\delta_t|, 1) \|e_t\|_1)} \quad (5)$$

and updates by $\alpha_t \delta_t e_t$. Our AdaptiveObGD variant maintains

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2)(\delta_t e_t)^2, \quad \rho_t = \frac{e_t}{\sqrt{\hat{v}_t} + \epsilon}, \quad (6)$$

then substitutes ρ_t for e_t in the ObGD bound and update. Bias correction in $\hat{v}_t$ is a local implementation choice. The method has no Adam first moment and differs from Adaptive Q(λ), which estimates the second moment of the current value gradient, clips δ_t , and does not use the ObGD bound (Gogianu et al., 2026).

3.2. Training Recurrent Memory

Exact RTU-RTRL. For a generic recurrence, the forward sensitivity is

$$J_t = \frac{\partial h_t}{\partial \theta} = \frac{\partial f_\theta}{\partial \theta} + \frac{\partial f_\theta}{\partial h_{t-1}} J_{t-1}. \quad (7)$$

Dense GRU/LSTM recurrence makes J_t costly. RTU’s diagonal complex-valued transition removes most cross-unit derivatives while retaining multiple learned time scales, making the compact trace exact (Elelimy et al., 2024).

One-step and true TBPTT(5). One-step training detaches h_{t-1} before every update, retaining only the current local derivative. True TBPTT(5) recomputes the newest prediction over a five-transition window from a detached boundary carry. Actor-critic eligibility traces remain active in both cases; “BPTT-1” refers only to recurrent-state derivatives.

Approximate e-prop. For recurrent parameter θ_i , our cell carries a local forward trace

$$\epsilon_{i,t} = \rho \epsilon_{i,t-1} + \left[\frac{\partial h_t}{\partial \theta_i} \right]_{\text{local}}, \quad \hat{g}_{i,t} = L_t \epsilon_{i,t}, \quad (8)$$

where L_t is the current readout learning signal. Symmetric feedback uses the current readout Jacobian; random feedback replaces it with a fixed matrix. Unlike BPTT-1, e-prop does *not* discard earlier contributions: it updates every transition while $\epsilon_{i,t}$ carries them forward. Our shared scalar $\rho = 0.9$ replaces the cell-specific GRU/LSTM recursions, so we call the method *approximate e-prop*, never “one-step e-prop” (Bellec et al., 2020).

Delta-rule memory. The delta baseline maps the current embedding to normalized query/key vectors q_t, k_t , value v_t , write rate β_t , and decay $d_t \in [0.90, 0.999]$. Its 16×64 fast-weight matrix is updated by

$$\hat{v}_t = M_{t-1}^\top k_t, \quad (9)$$

$$M_t = d_t M_{t-1} + \beta_t k_t (v_t - \hat{v}_t)^\top, \quad (10)$$

$$h_t = \text{LayerNorm}(M_t^\top q_t). \quad (11)$$

The residual write corrects the value already stored at k_t , following DeltaNet (Schlag et al., 2021); learned decay borrows the forgetting principle of Gated DeltaNet (Yang et al., 2025). Training detaches M_{t-1} before each parameter update, making this a one-step-gradient baseline. It is neither exact RTRL nor the full parallel Gated DeltaNet architecture.

3.3. Batched and Fixed-Memory Controls

PPO-GRU uses 128-step rollouts, generalized advantage estimation, four epochs, and a clipped surrogate (Schulman et al., 2016; 2017). We set `num_envs = 1` in the RepeatPrevious comparison, but PPO still performs batched multi-epoch optimization. PPO-RS models $p_\theta(h_{t+1} | o_t, h_t)\pi_\theta(a_t | o_t, h_{t+1})$ and applies PPO to the joint log-probability. It avoids BPTT through sampled states but requires complete rollouts, so it is a control for memory learning rather than a strict streaming method.

The frozen-SSM control uses fixed zero-preserving observation features and stable diagonal multiscale dynamics; only actor and critic readouts are learned. An external checkpoint interface is implemented, but the reported pilot uses a deterministic structured prior, not pretrained language-model weights.

4. Experimental Setup

Table 1. Experimental stages. Budgets are per seed. “Mixed” denotes archived streaming runs with both ObGD variants; PPO uses Adam throughout.

Stage	Budget	Seeds	Stream update	Purpose
CartPole	100k steps	3	ObGD	update check
Passive T-Maze	100k ep.	3	ObGD	memory horizon
POPGym archive	170k steps	≤ 3	mixed	task screen
RepeatPrevious	168–170k	3	AdaptiveObGD	method comparison
Stable RepeatPrevious	204.8k	3	ObGD	stability study
Frozen SSM	20k steps	3	ObGD	fixed memory
CT-graph	102.4k	5	ObGD	adaptation
Hidden-state probes	256 ep.	3	—	representation

Environments. CartPole-v1 is the fully observed optimization check (Towers et al., 2024). Passive T-Maze follows Memory-RL (Ni et al., 2022): a binary cue precedes a corridor and a terminal arm choice; fixed-arm success is 0.5. We train at $L \in \{2, 3, 5, 10, 15, 30, 50\}$ and evaluate 1,024 deterministic episodes per seed. Active T-Maze additionally requires visiting the cue and returning to the junction.

The POPGym screen covers Autoencode, Battleship, Concentration, and RepeatPrevious at three difficulties (Morad et al., 2023). We deduplicate the archive to 134 runs and compare Easy tasks at a common 170k-transition horizon. The focused RepeatPrevious study uses seeds 0, 42, 123 and the final 500 episodes; the stability study evaluates three

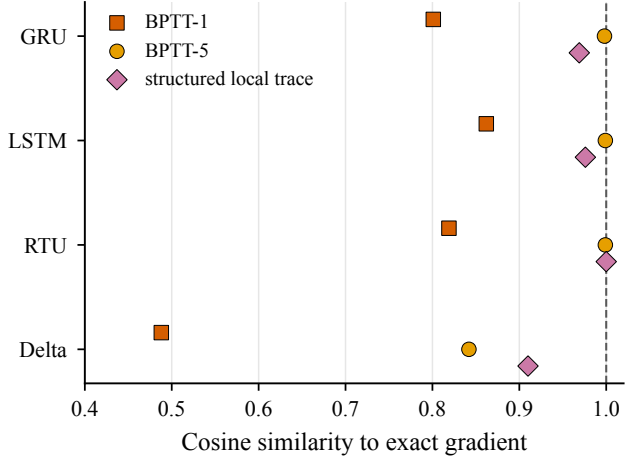


Figure 1. Cosine similarity to the exact horizon-25 recurrent gradient, averaged over eight synthetic seeds. BPTT(5) nearly recovers the exact GRU, LSTM, and RTU gradients. “Structured local trace” is the cell-local forward approximation.

204.8k-transition checkpoints on 256 episodes each.

The continuing CT-graph has branching factor 2, depth 2, and a rewarding leaf that switches every 128 transitions (Soltoggio et al., 2023). PPO-GRU and StreamAC-GRU run for 102.4k transitions on five seeds. Adaptation gain is late-minus-early return within a phase. Representation analysis uses episode-disjoint ridge probes and held-out k-means; DelayedCue details appear in the appendix.

Statistics. Bars and tables report mean $\pm$ sample SD over seeds. POPGym point plots summarize the final 20% of the common horizon; the RepeatPrevious curve uses a 100-episode trailing mean. Error bands and bars are descriptive percentile-bootstrap intervals. We make no significance claim with three to five seeds.

5. Results

5.1. Implementation Audit Changes the Baselines

The archived `tbptt_steps=5` option controlled compiler unrolling while stopping the recurrent carry at every step. Across 174 paired configurations, nominal one- and five-step variants differed by only 0.0064 in final score. We replaced it with a differentiated five-transition window. The symmetric e-prop branch also bypassed its accumulated trace; the corrected path combines the current head Jacobian with that trace. Method claims use corrected runs.

In Figure 1, BPTT(5) reaches cosine similarity 0.998–0.999 to the exact GRU, LSTM, and RTU gradients; BPTT(1) reaches 0.801–0.862. For the delta cell, BPTT(5) reaches

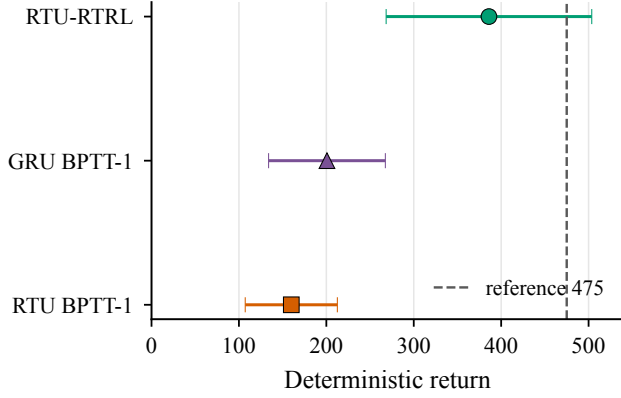


Figure 2. CartPole-v1 frozen-policy return after 100k transitions. Points show means and error bars show sample SD over three seeds; the dashed line marks the reference return of 475.

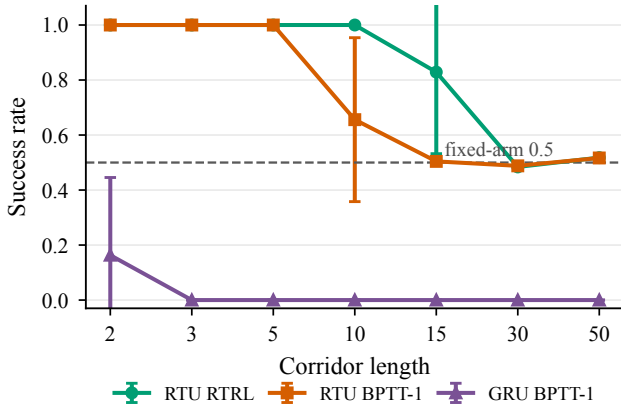


Figure 3. Passive T-Maze success after 100k training episodes at each corridor length. Curves show means and sample SD over three seeds. The dashed line is the 0.5 fixed-arm rate.

0.842 and the local trace 0.910.

5.2. RTRL Helps Memory, but Not Indefinitely

On CartPole, RTU-RTRL reaches 386.0 ± 117.6 , against 200.7 ± 66.8 for GRU BPTT-1 and 159.9 ± 52.7 for RTU BPTT-1 (Figure 2). Only one RTU-RTRL seed exceeds 475.

Passive T-Maze separates the estimators more clearly (Figure 3). RTU-RTRL solves $L \leq 10$ and reaches 0.829 ± 0.297 at $L = 15$. RTU BPTT-1 is reliable through $L = 5$ and returns to the fixed-arm rate by $L = 15$; both RTUs obtain about 0.5 at $L = 30-50$. GRU BPTT-1 rarely makes a valid terminal choice for $L \geq 3$. Reported runs use a dense input projection after an earlier sparse initializer erased both input dimensions.

PPO-RS trained at $L = 5$ reaches deterministic success 1.0

through $L = 50$; stochastic execution reaches 0.839 ± 0.126 at $L = 50$. Its complete-rollout GAE makes it a memory control, not a strict streaming method.

5.3. PPO Dominates the Short POPGym Comparison

Autoencode. Every method finishes near -0.5 (Figure 4); their intervals overlap and no run develops a sustained learning trend.

Battleship. RTU-RTRL, PPO-GRU, GRU BPTT-1, and e-prop RNN finish between -0.46 and -0.45 (Figure 5); corrected e-prop GRU reaches -0.532 ± 0.023 .

Concentration. RTU-RTRL and GRU BPTT-1 reach approximately -0.22 , ahead of PPO at this budget (Figure 6). Corrected e-prop RNN has two complete seeds. All returns remain negative.

RepeatPrevious. RepeatPrevious is the only task in the screen with a near-solved policy (Figure 7). PPO-GRU reaches 0.994 ± 0.001 over the final 500 episodes. RTU-RTRL reaches 0.697 ± 0.036 , RTU BPTT-1 0.377 ± 0.059 , and corrected matched RTU TBPTT(5) 0.616 ± 0.036 . Extending the recurrent gradient improves return by 0.239 over BPTT(1) and leaves a 0.081 gap to exact RTU-RTRL. PPO remains 0.297 above RTU-RTRL.

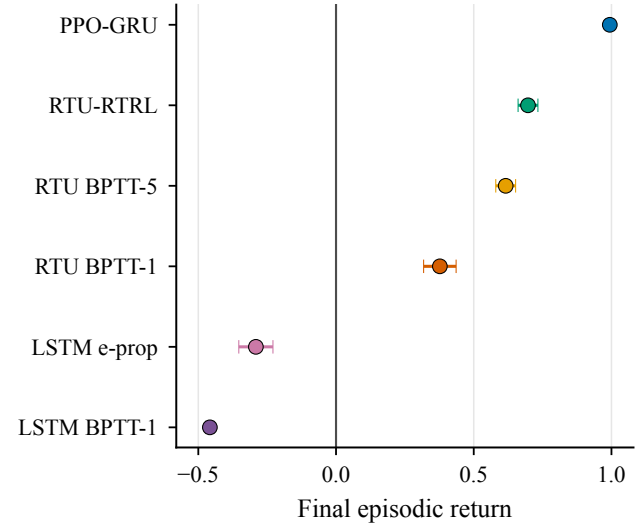


Figure 8. Final RepeatPrevious Easy return for the focused comparison. Points show means and error bars show sample SD over three seeds.

Approximate e-prop improves LSTM return from -0.458 ± 0.014 to -0.291 ± 0.062 ; GRU e-prop reaches -0.318 ± 0.019 and the delta cell -0.458 ± 0.038 (Figure 8). None learns the task.

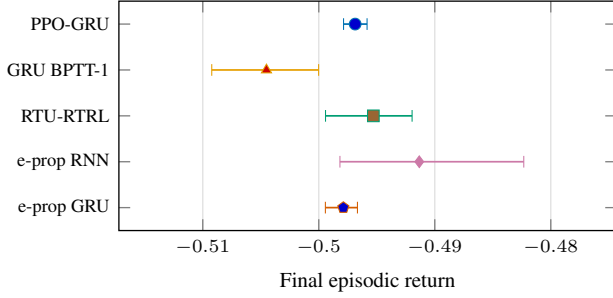


Figure 4. Autoencode Easy final return over the last 20% of the common 170k-transition horizon. Points show means over three seeds; bars are descriptive percentile-bootstrap intervals. The narrow axis exposes overlap among methods.

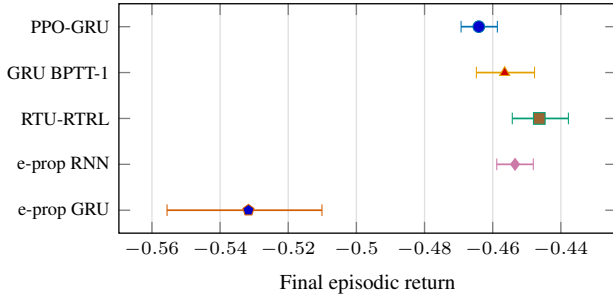


Figure 5. Battleship Easy final return at 170k transitions. Aggregation matches Figure 4; higher is better.

5.4. Constraining RTU Dynamics Improves RepeatPrevious

Table 2. Frozen deterministic evaluation on RepeatPrevious Easy after 204.8k training transitions. Entries are mean $\pm$ SD over three seeds and 256 evaluation episodes per seed.

Method	Optimizer	Return	Valid-action accuracy	Target probe
PPO-GRU	Adam	0.998 ± 0.003	0.999 ± 0.001	0.999 ± 0.001
StreamAC-GRU	ObGD	-0.083 ± 0.022	0.459 ± 0.011	0.644 ± 0.007
StreamAC-RTU	ObGD	0.904 ± 0.063	0.952 ± 0.031	1.000 ± 0.000

Bounding RTU radii at $r_{\max} = 0.99$ avoids the NaN seen in one unconstrained pilot. All three seeds learn, with frozen return 0.904 ± 0.063 (Table 2). Capacity, optimizer, budget, and dynamics also differ from the 170k study, so the gain cannot be assigned to the radius bound alone.

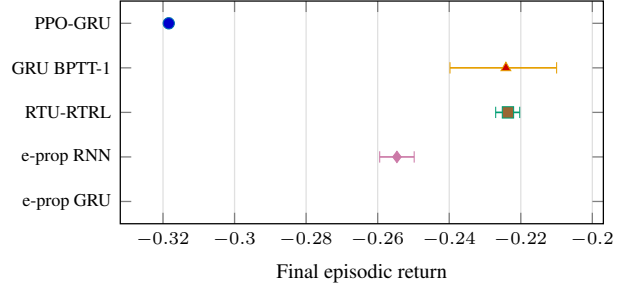


Figure 6. Concentration Easy final return at 170k transitions. Corrected e-prop RNN has two complete seeds; other methods have three. Bars are descriptive percentile-bootstrap intervals.

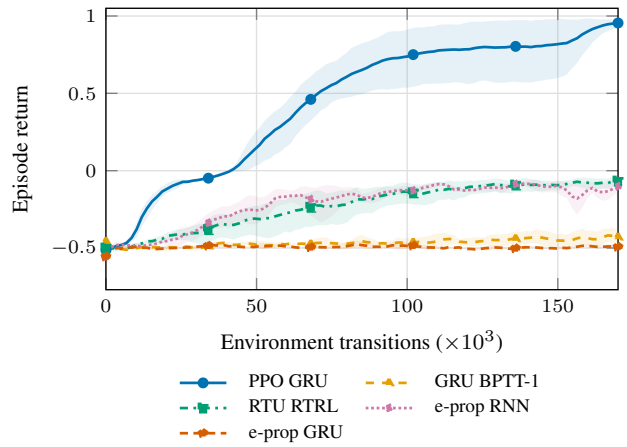


Figure 7. RepeatPrevious Easy at 170k transitions. Lines show three-seed means after a 100-episode trailing average; bands are descriptive percentile-bootstrap intervals.

5.5. Frozen Memory and Active Control Remain Unsolved

Table 3. Harder controls. Success near 0.5 is a fixed-choice policy, not cue-conditioned behavior.

Environment	Method	Optimizer	Seeds	Success
Passive $L = 10$	PPO + frozen SSM	Adam	3	0.000 ± 0.000
	PPO-GRU	Adam	3	0.497 ± 0.028
	StreamAC + frozen SSM	ObGD	3	0.335 ± 0.291
	StreamAC-GRU	ObGD	3	0.501 ± 0.017
Active $L = 5$	PPO-GRU	Adam	5	0.488 ± 0.029
	StreamAC-GRU	ObGD	5	0.505 ± 0.027
	StreamAC-LSTM	ObGD	5	0.510 ± 0.020

The frozen SSM preserves the initial Passive T-Maze cue under a held-out linear probe, but the learned readouts do not use it reliably (Table 3). Active T-Maze also remains at the fixed-choice rate for the five-seed PPO and streaming

configurations. Cue retention is therefore insufficient to explain policy failure in these runs.

5.6. Hidden States Encode Selective Working Memory

Table 4. RepeatPrevious Easy representation diagnostics for the same frozen checkpoints as Table 2. Linear probes use episode-disjoint splits. Homogeneity measures whether unsupervised k-means clusters align with the hidden target.

Method	Target probe	Target homogeneity	Return
PPO-GRU	0.999 ± 0.001	0.488 ± 0.030	0.998 ± 0.003
StreamAC-GRU	0.644 ± 0.007	0.039 ± 0.004	-0.083 ± 0.022
StreamAC-RTU	1.000 ± 0.000	0.002 ± 0.001	0.904 ± 0.063

The successful RTU stores the target in a linearly accessible but non-clustered code (Table 4); k-means homogeneity therefore misses usable memory. Four-step observation history also decodes the target perfectly, so the probe demonstrates compression rather than new information. On noisy DelayedCue, a trained one-layer LSTM reaches cue-probe accuracy 1.000, against 0.489 for raw observations and 0.832 for an untrained LSTM. Cue-cluster ARI rises from 0.024 to 0.919. In a two-layer LSTM, cue-distance alignment is stronger in the last layer ($\rho = 0.853$ versus 0.201); the appendix reports the full single-checkpoint diagnostic.

5.7. Lifelong CT-Graph Shows No Reliable Adaptation

Table 5. Continuing CT-graph, depth 2, reward switch every 128 transitions. Mean $\pm$ SD over five seeds. A fixed leaf has expected return 0.5.

Method	Optimizer	Final return	Success	Adaptation gain
PPO-GRU	Adam	0.495 ± 0.012	0.243 ± 0.012	0.003 ± 0.007
StreamAC-GRU	ObGD	0.489 ± 0.013	0.260 ± 0.009	0.015 ± 0.010

Both agents remain near the 0.5 fixed-leaf return (Table 5). StreamAC’s within-phase gain is 0.015 ± 0.010 but does not improve final return. Immediate updates therefore show no reliable adaptation in this configuration.

6. Discussion

Gradient quality depends on recurrent dynamics. RTU-RTRL makes exact forward sensitivities practical by restricting recurrence: at width 64, its compact trace uses roughly 0.063 MiB, compared with 6.05 MiB for a dense GRU block and 16.13 MiB for an LSTM block. TBPTT(5) nearly recovers the exact dense-cell gradient, while bounded RTU radii prevent catastrophic dynamics in the three reported seeds. The stability experiment changes several factors, however, and does not isolate the radius constraint.

PPO remains the stronger matched baseline. PPO-GRU leads RepeatPrevious, consistent with tuned recurrent baselines (Ni et al., 2022). The separate stabilized RTU run narrows the gap, but changes capacity, optimizer, and budget. PPO also learns little on several short-budget POPGym tasks and fails Active T-Maze. The comparison is task- and budget-dependent; a single Concentration seed improves only after 7.9M transitions.

Optimizer evidence is not factorial. Ordinary ObGD is used in CartPole, T-Maze, stable RepeatPrevious, and CT-graph, while the matched 170k study uses AdaptiveObGD. Archived POPGym runs mix both variants. The results establish that both optimizers can support learning in some configurations, but they do not estimate an ObGD–AdaptiveObGD treatment effect.

6.1. Where Streaming Memory Helped

The experiments contain several reproducible positive results, although none supports a blanket advantage for streaming updates. Table 6 collects the cases in which recurrent state or a longer online credit-assignment path produced a measurable benefit. The comparisons are deliberately local: each claim uses runs from the same environment and evaluation protocol, and the final column records the remaining confound or stronger baseline.

Table 6. Positive evidence in the audited experiments. Returns and success rates are mean $\pm$ SD; all comparisons use three seeds. “One-step” denotes a detached incoming recurrent state.

Setting	Positive result	Local comparator	Qualification
CartPole-v1	RTU-RTRL 386.0 ± 117.6	GRU one-step 200.7 ± 66.8	reference is 475
Passive T-Maze	RTU-RTRL solves $L \leq 10$	one-step fails beyond $L = 5$	degrades by $L = 30$
RepeatPrevious	RTU-RTRL 0.697 ± 0.036	RTU one-step 0.377 ± 0.059	PPO reaches 0.994
RepeatPrevious	RTU TBPTT(5) 0.616 ± 0.036	RTU one-step 0.377 ± 0.059	five-step buffer
Stable RTU	frozen return 0.904 ± 0.063	StreamAC-GRU -0.083 ± 0.022	changed optimizer/capacity

Exact online credit improves over a one-step recurrent surrogate. On CartPole, RTU-RTRL improves the frozen-policy return by 185.3 over GRU BPTT-1 and by 226.1 over RTU BPTT-1. The uncertainty is large, but the direction agrees across the three seeds and the exact method is the only recurrent-gradient variant to exceed the reference return in one seed. Passive T-Maze gives the cleaner memory result: RTU-RTRL maintains success 1.0 through corridor length 10 and reaches 0.829 ± 0.297 at length 15, whereas RTU BPTT-1 is reliable only through length 5. Exact traces therefore extend the usable credit horizon in a controlled delayed-cue problem, even though they do not prevent failure at lengths 30–50.

A short differentiated window recovers most of the exact-gradient gain. The corrected five-transition RTU window raises RepeatPrevious return from 0.377 ± 0.059 to 0.616 ± 0.036 . It closes 75% of the 0.320 gap between

one-step training and exact RTU-RTRL, while storing only five transitions. The corresponding gradient diagnostic is consistent with the behavioral result: TBPTT(5) is much closer to the exact recurrent gradient than BPTT(1). A small buffer is therefore a strong practical configuration when strict transition-wise updates are not required. The remaining 0.081 gap to RTU-RTRL shows that the longer dependency is not fully captured.

Stable recurrent dynamics can turn accessible memory into control. The bounded-radius RTU run is the strongest streaming result in the study. Its frozen return is 0.904 ± 0.063 , valid-action accuracy is 0.952 ± 0.031 , and target-probe accuracy is 1.000. All three seeds learn. By contrast, the matched probe for StreamAC-GRU is only 0.644 ± 0.007 and its return is negative. The result demonstrates that a streaming actor-critic can solve RepeatPrevious with a recurrent state trained online. It does not identify a single causal ingredient because the successful configuration also changes width, optimizer, budget, and the radius constraint. The defensible conclusion is configurational: bounded dynamics, an exact structured trace, and ObGD are jointly sufficient in this setting.

DelayedCue links learned state geometry to successful behavior. The one-layer LSTM obtains return 5.860/6 after 200k transitions and linearly exposes the cue with accuracy 1.000 on cue-free observations. Raw observations give 0.489, while an untrained LSTM gives 0.832. Cue-cluster ARI rises from 0.024 to 0.919, and the two-layer model concentrates cue-distance geometry in its last layer. Unlike the frozen-SSM pilot, this checkpoint combines decodable memory with near-optimal control. The analysis is limited to one checkpoint per architecture, but it supplies a concrete positive example in which the learned hidden state is both interpretable and behaviorally useful.

Stored information need not control behavior. The frozen SSM preserves the cue while its readout fails, whereas the successful RTU uses a linearly decodable but weakly clustered code. PPO-RS learns long memory from trajectory scores but requires rollout batches. The CT-graph adds a separate negative result: immediate updates do not solve interference or inference of the current reward location. Memory, credit assignment, and control require separate measurements (Ni et al., 2023).

7. Conclusion

RTU-RTRL extends the Passive T-Maze horizon and beats one-step training on RepeatPrevious, but matched PPO-GRU remains stronger. TBPTT(5) recovers most of the exact gradient and 75% of the one-step-to-RTRL return gap with a five-transition buffer. A bounded RTU trained

with ObGD reaches 0.904 ± 0.063 frozen return across three seeds, establishing that transition-wise recurrent actor-critic learning can solve RepeatPrevious in a suitable configuration. DelayedCue provides a second positive case: near-optimal return coincides with a linearly decodable and strongly clustered cue representation. Hidden-state probes also show why memory accuracy alone is not enough; the frozen SSM retains its cue but fails to control reliably. E-prop, delta-rule, short-budget POPGym, active T-Maze, and CT-graph configurations show no reliable behavioral gain. Exact or longer online gradients are useful when the recurrent dynamics remain stable and the task exposes a learnable memory signal; optimizer, capacity, and budget determine whether that advantage becomes behavior.

References

- Al-Hafez, F., Zhao, G., Peters, J., and Tateo, D. Time-efficient reinforcement learning with stochastic stateful policies. *arXiv preprint arXiv:2311.04082*, 2023. URL <https://arxiv.org/abs/2311.04082>.
- Bellec, G., Scherr, F., Subramoney, A., Hajek, E., Salaj, D., Legenstein, R., and Maass, W. A solution to the learning dilemma for recurrent networks of spiking neurons. *Nature Communications*, 11(1):3625, 2020. URL <https://doi.org/10.1038/s41467-020-17236-y>.
- Cho, K., van Merriënboer, B., Gulcehre, C., Bahdanau, D., Bougares, F., Schwenk, H., and Bengio, Y. Learning phrase representations using RNN encoder-decoder for statistical machine translation. *arXiv preprint arXiv:1406.1078*, 2014. URL <https://arxiv.org/abs/1406.1078>.
- Elelimy, E., White, A., Bowling, M., and White, M. Real-time recurrent learning using trace units in reinforcement learning. In *Advances in Neural Information Processing Systems*, 2024. URL <https://arxiv.org/abs/2409.01449>.
- Elsayed, M., Vasan, G., and Mahmood, A. R. Streaming deep reinforcement learning finally works. *arXiv preprint arXiv:2410.14606*, 2024. URL <https://arxiv.org/abs/2410.14606>.
- Farr, N., Reddi, A., D’Eramo, C., and Peters, J. Streaming reinforcement learning under partial observability with real-time recurrent learning. *arXiv preprint arXiv:2605.24709*, 2026. URL <https://arxiv.org/abs/2605.24709>.
- Gogianu, F., Lutu, A. C., and Pascanu, R. Revisiting adam for streaming reinforcement learning. *arXiv preprint arXiv:2605.06764*, 2026. URL <https://arxiv.org/abs/2605.06764>.
- Hochreiter, S. and Schmidhuber, J. Long short-term memory. *Neural Computation*, 9(8):1735–1780, 1997. URL <https://doi.org/10.1162/neco.1997.9.8.1735>.
- Morad, S., Kortvelesy, R., Bettini, M., Liwicki, S., and Prorok, A. POPGym: Benchmarking partially observable reinforcement learning. *arXiv preprint arXiv:2303.01859*, 2023. URL <https://arxiv.org/abs/2303.01859>.
- Ni, T., Eysenbach, B., and Salakhutdinov, R. Recurrent model-free RL can be a strong baseline for many POMDPs. In *Proceedings of the 39th International Conference on Machine Learning*, pp. 16691–16723, 2022. URL <https://arxiv.org/abs/2110.05038>.
- Ni, T., Ma, M., Eysenbach, B., and Bacon, P.-L. When do transformers shine in RL? decoupling memory from credit assignment. In *Advances in Neural Information Processing Systems*, 2023. URL <https://arxiv.org/abs/2307.03864>.
- Paischer, F., Adler, T., Patil, V., Bitto-Nemling, A., Eghbalzadeh, H., and Hochreiter, S. History compression via language models in reinforcement learning. *arXiv preprint arXiv:2205.12258*, 2022.
- Schlag, I., Irie, K., and Schmidhuber, J. Linear transformers are secretly fast weight programmers. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139, pp. 9355–9366. PMLR, 2021. URL <https://proceedings.mlr.press/v139/schlag21a.html>.
- Schulman, J., Moritz, P., Levine, S., Jordan, M., and Abbeel, P. High-dimensional continuous control using generalized advantage estimation. In *International Conference on Learning Representations*, 2016. URL <https://arxiv.org/abs/1506.02438>.
- Schulman, J., Wolski, F., Dhariwal, P., Radford, A., and Klimov, O. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017. URL <https://arxiv.org/abs/1707.06347>.
- Soltoggio, A., Ben-Iwhiwhu, E., Peridis, C., Ladosz, P., Dick, J., Pilly, P. K., and Kolouri, S. The configurable tree graph: Measurable problems in partially observable and distal reward environments for lifelong reinforcement learning. *arXiv preprint arXiv:2302.10887*, 2023. URL <https://arxiv.org/abs/2302.10887>.
- Towers, M., Kwiatkowski, A., Terry, J., Balis, J. U., De Cola, G., Deleu, T., Goulão, M., Kallinteris, A., Krimmel, M., KG, A., et al. Gymnasium: A standard interface for reinforcement learning environments. *arXiv preprint arXiv:2407.17032*, 2024. URL <https://arxiv.org/abs/2407.17032>.
- Williams, R. J. and Zipser, D. A learning algorithm for continually running fully recurrent neural networks. *Neural Computation*, 1(2):270–280, 1989. URL <https://doi.org/10.1162/neco.1989.1.2.270>.
- Yang, S., Kautz, J., and Hatamizadeh, A. Gated delta networks: Improving Mamba2 with delta rule. In *International Conference on Learning Representations*, 2025. URL <https://arxiv.org/abs/2412.06464>.

A. Detailed Hyperparameters

Table 7. Main recurrent actor–critic configurations. Actor and critic have separate parameters and recurrent states.

Method	Backbone	Recurrent gradient	Optimizer/update	Key parameters
PPO-GRU	embed 64, GRU 64	rollout BPTT	Adam, PPO clip	$\gamma = .99$, GAE $\lambda = .95$, $\text{lr } 7 \times 10^{-4}$
RTU-RTRL	embed 64, RTU 128 real outputs	compact exact RTRL	AdaptiveObGD	$\lambda = .97$, $\text{lr } 1$, $\kappa_\pi = .2$, $\kappa_V = .5$
Stable RTU-RTRL	embed 64, RTU-64 complex state	compact exact RTRL	ObGD	$r \in [0, .99]$, $\lambda = .9$, $\text{lr } 1$
RTU BPTT-1	same RTU	detached incoming carry	AdaptiveObGD	same as RTU-RTRL
RTU BPTT-5	same RTU	true five-step window	AdaptiveObGD	matched $\lambda = .97$
GRU/LSTM e-prop	cell 64, LayerNorm	local trace, symmetric signal	AdaptiveObGD	trace decay .9, sparsity .9
Delta fast-weight cell	fast weights 16×64	BPTT-1; detached M_{t-1}	AdaptiveObGD	normalized q/k , $d_t \in [.90, .999]$

The article-faithful long POPGym StreamAC uses Dense(64)–LayerNorm–LeakyReLU–RTU(192)–head, separate actor and critic RTUs, 90% sparse initialization, $\gamma = 0.99$, $\lambda = 0.8$, entropy coefficient 0.095, learning rate 1.0, and ObGD $\kappa = 3/2$. Every transition is used once; no replay, rollout, BPTT, or target network is present. Autoencode uses seeds 0–2 and 5M transitions per seed. The Concentration snapshot uses seed 0 and is frozen at 17.2M transitions.

The AdaptiveObGD pilot adds $\beta_2 = 0.999$ and $\epsilon = 0.1$ to the same architecture. A 10k-transition stability sweep over $\epsilon \in \{10^{-8}, 0.01, 0.1\}$ kept every run finite but did not distinguish returns, so 0.1 was selected from the optimizer-motivated hypothesis of Gogianu et al. (2026), not from pilot reward.

The continuing CT-graph comparison uses one environment, 102,400 transitions, five seeds, branching 2, depth 2, linear leaf rewards, and a 128-transition switch interval. PPO uses 128-step rollouts, one minibatch, and four epochs; StreamAC uses a two-layer GRU with direct entropy updates. DelayedCue has delay 5, eight Gaussian nuisance dimensions, dense binary-report rewards, and maximum return 6. Frozen representation analysis uses episode-disjoint train/test folds.

B. Exploratory Long-Budget Snapshots

Table 8. Long-budget observations excluded from main comparative claims. No learning curve is shown because Concentration has one seed and the Autoencode effect is negligible.

Environment	Budget	Seeds	Final 250k mean	Interpretation
Autoencode Easy	5M	3	approximately -0.493	finite, reproducible, near random regime
Concentration Easy	17.2M	1	0.120	late improvement; exploratory only
AdaptiveObGD Autoencode	3.45M snapshot	1	approximately -0.49	incomplete; excluded

The Concentration seed moves from a 250k-transition mean of -0.048 at 5M to sustained positive return after roughly 7.9M transitions. The observation motivates a pre-registered multi-seed long-budget run but cannot establish reproducibility, uncertainty, or superiority. The corresponding per-seed graph from the working report is intentionally omitted.

C. Additional POPGym Results

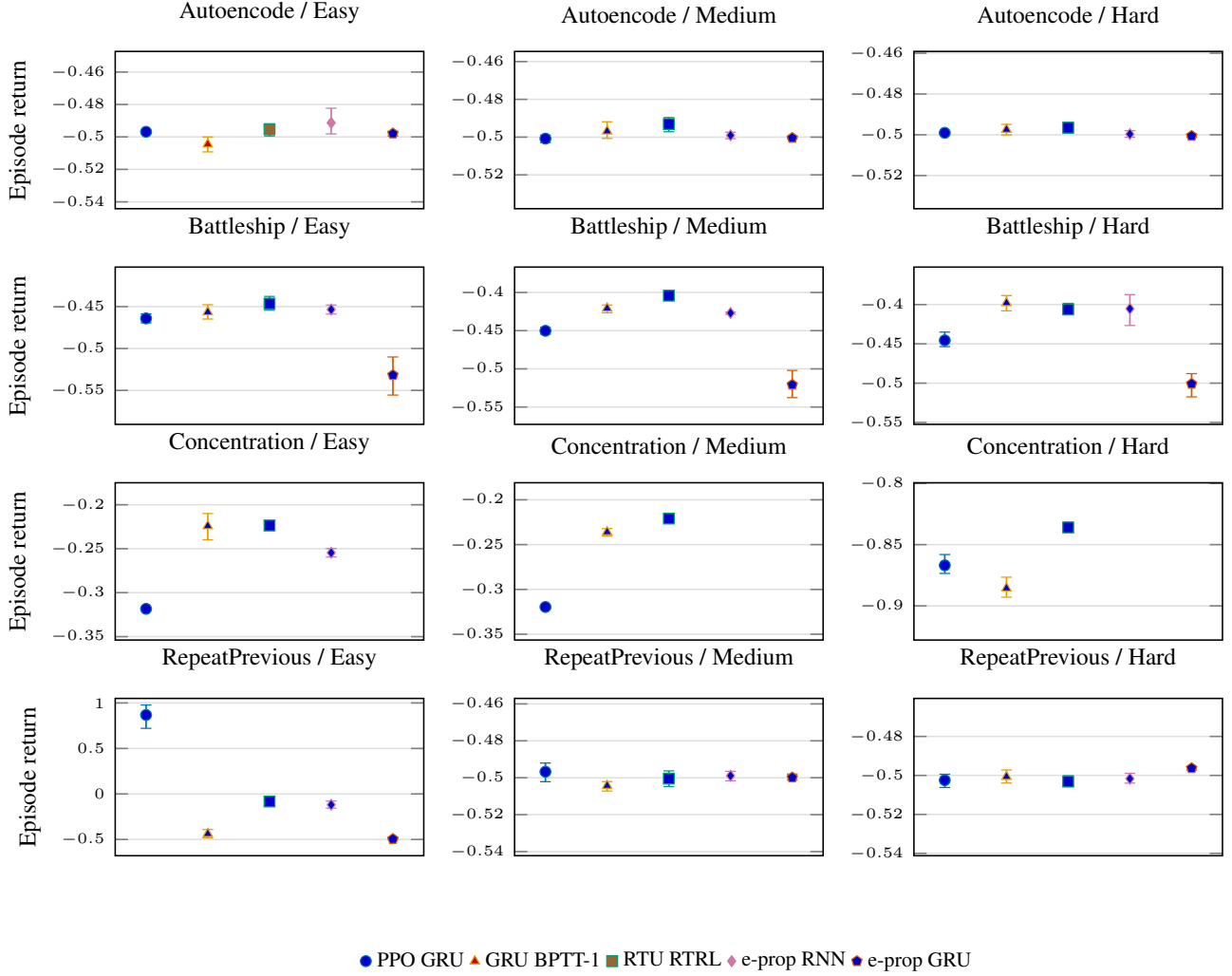


Figure 9. Final POPGym performance over the final 20% of the common 170k-transition horizon. Points are seed means; error bars are descriptive percentile-bootstrap intervals over up to three seeds. Axis ranges differ by panel and are printed explicitly. Missing markers indicate missing complete comparable runs.

No cross-task ranking should be inferred from absolute return. At this budget, RepeatPrevious Easy is the only task with a near-solved PPO policy. Medium and Hard variants remain near their low-return regimes for all methods. Small differences on Battleship and Concentration motivate longer training but are not evidence of solved memory.

D. Long-Horizon T-Maze Stress Test

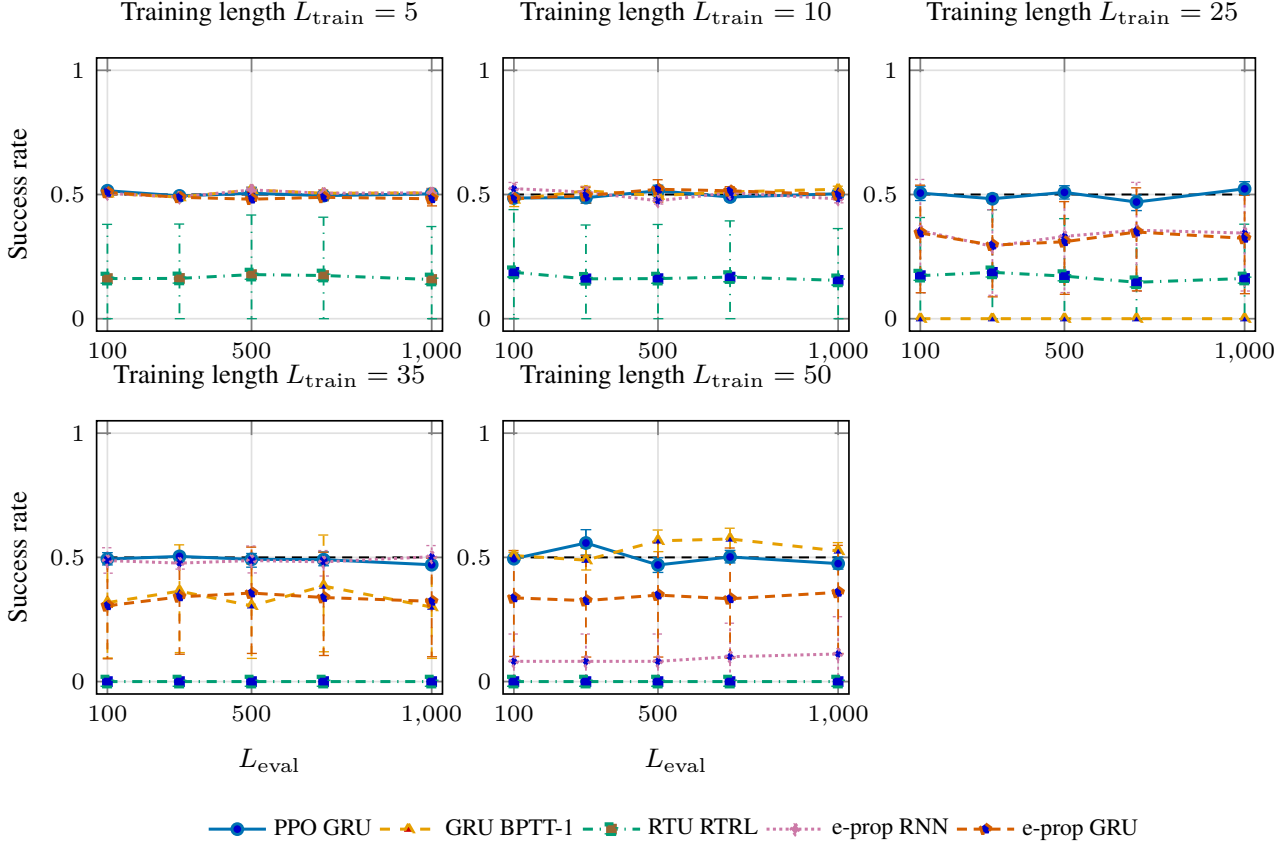


Figure 10. Frozen long-horizon T-Maze evaluation from the audited archive. Each panel fixes the training corridor length and evaluates corridors from 100 to 1,000. Points average three seeds; the dashed line is the 0.5 fixed-arm reference. None of the compared methods demonstrates reliable cue-conditioned transfer at these horizons.

Several curves fall below 0.5 because a policy may fail to reach the junction or may select invalid actions; zero is therefore not “worse memory” in isolation. The stress test is most safely interpreted as failure to establish reliable cue-dependent behavior, not as a calibrated ranking below chance.

E. Frozen SSM Pilot

The passive T-Maze corridor length is 10, the comparison horizon is 20k transitions, and the final window contains 500 completed episodes per seed. The fixed encoder is zero-preserving so corridor zeros do not continually overwrite the initial cue. Actor and critic readouts are the only trainable weights. A separate linear probe decodes the cue at the junction with 100% held-out accuracy and class-mean distance 0.476, yet no RL method exceeds the fixed-arm reference reproducibly. The failed optimization therefore cannot be attributed to complete cue erasure.

F. DelayedCue State Geometry

After 200k transitions, a one-layer LSTM reaches 5.860/6. On 512 held-out episodes, cue-probe accuracy on cue-free observations is 1.000, versus 0.489 for raw observations and 0.832 for an untrained LSTM; cue-cluster ARI rises from 0.024 to 0.919. In a two-layer LSTM, cue-distance alignment is stronger in the last layer ($\rho = 0.853$ versus 0.201), whereas time-distance alignment is stronger in the first (0.550 versus 0.082).